BIO 304 . HUMAN ANATOMY & PHYSIOLOGY . MEDMASTERS COLLABORATIVE

Module 2: Chemistry of Life

Week 2 teaching guide . 3 topics

This is the instructor-facing companion to the student pre-work hub. For every topic, you have the science to teach, the recommended approach to learning that material, and why it matters to your students' future patients. Lecture order, common misconceptions, and pre-video drawing prompts are baked in so each video lesson can hit the same pedagogical beats.

Topics in this module

1. Atoms, Bonds, and Water

Subatomic particles, ionic vs covalent vs hydrogen bonds, and why water is the medium of life.

2. Biological Macromolecules

The four major polymer classes: carbohydrates, lipids, proteins, and nucleic acids.

3. Enzymes and ATP

How enzymes lower activation energy, what regulates them, and the role of ATP as the cell's energy currency.

Module 2 of 17 . Dr. Sharilyn Rennie

TOPIC 1 OF 3 . WEEK 2 . 7 CARDS (4 DOK 1 / 2 DOK 2 / 1 DOK 3)

Atoms, Bonds, and Water

Subatomic particles, ionic vs covalent vs hydrogen bonds, and why water is the medium of life.

The Science

Chemistry is the substrate the rest of the course rides on, even when it does not look chemical. Lecture density should be low. Pick three things to nail: subatomic structure, the three bond types ranked by strength and behavior, and the properties of water.

Bonds: ionic (full electron transfer, strong in dry, weak in water because water screens the charges), covalent (shared electrons, very strong, can be polar or nonpolar based on electronegativity), hydrogen (weak attraction between a partial-positive H and a partial-negative O or N, weak individually but enormously powerful in aggregate). The hydrogen bond is the bond that holds proteins folded, DNA strands paired, and water behaving like a liquid at room temperature. Spend time here.

Water is the medium of life and behaves the way it does because it is polar and small. Polarity makes it a universal biological solvent (anything with a charge or partial charge dissolves; anything hydrophobic does not, which is why membranes form). High specific heat buffers the body against temperature swings. High heat of vaporization is why sweating cools effectively. The chemistry is the reason your patient can stand in a hot kitchen all day without their core temperature climbing.

Teaching This Topic

Before the video (drawing prompt)

Have students draw a water molecule with partial charges labeled, then a second water molecule hydrogen-bonded to it. They have to commit to where the bond lives before the answer is shown.

Common misconceptions to address

- Students think hydrogen bonds are intramolecular (within one molecule). In water and most biology they are intermolecular (between molecules).
- Polar and ionic get conflated. Polar is a partial charge; ionic is a full one.
- Students assume strong bonds matter more than weak ones. In biology, weak bonds matter MORE because they are reversible: that is the basis of enzyme-substrate binding, receptor-ligand binding, and DNA base pairing.

Order of operations (what to teach first)

Atomic structure briefly, then bonds (ranked by strength), then water properties explained as consequences of the bonds. Lead with structure; consequences follow.

Self-test prompts (for students before the recall deck)

- Draw three water molecules hydrogen-bonded together.
- Without notes, list three properties of water and why each one matters for life.

- Rank ionic, covalent, and hydrogen bonds by strength.

Why This Matters to Your Patients

This shows up later when you wonder why an IV fluid is isotonic, why a fever destabilizes enzymes (hydrogen bonds breaking), why dehydration concentrates blood, and why anti-cancer drugs that disrupt hydrogen bonding (cisplatin and DNA crosslinks) cause specific kinds of cellular damage. Nurses titrating IV fluids are managing osmotic gradients, which are water-property problems. The chemistry is invisible until something goes wrong; then it becomes the whole story.

TOPIC 2 OF 3 . WEEK 2 . 7 CARDS (4 DOK 1 / 2 DOK 2 / 1 DOK 3)

Biological Macromolecules

The four major polymer classes: carbohydrates, lipids, proteins, and nucleic acids.

The Science

Four classes, four jobs. Carbohydrates for short-term energy. Lipids for long-term storage, membranes, and signaling molecules. Proteins for everything else (enzymes, structure, transport, recognition). Nucleic acids for information storage and transmission. Drive the class-to-function pairing hard; otherwise students just memorize structures with no organizing principle.

Polymers and monomers: students should be able to name the monomer of each class. Carbs: monosaccharides. Proteins: amino acids. Nucleic acids: nucleotides. Lipids do not technically polymerize from monomers, which is itself worth noting (this is a common test point). Triglycerides are glycerol plus three fatty acids; phospholipids are the same minus one fatty acid plus a phosphate head, and that single change is what lets them form bilayers.

Protein structure deserves its own beat. Primary (sequence) determines secondary (alpha helix and beta sheet patterns from hydrogen bonding), which determines tertiary (3D fold from R-group interactions), which sometimes assembles into quaternary (multiple chains). Function depends on tertiary and quaternary. Denature the fold (heat, pH) and you lose the function, even though every atom is still present. This explains fever's effect on enzymes, gastric acid's effect on dietary protein, and why egg whites turn solid when cooked.

Teaching This Topic

Before the video (drawing prompt)

Ask students to sketch one molecule from each class and label the monomer if applicable. They have to identify the building block before you show the structure.

Common misconceptions to address

- Lipids are not all the same. Triglycerides, phospholipids, and steroids look completely different and do different jobs.
- Students think 'denaturation' means destruction. It means unfolding; the chain is still there, just non-functional.
- DNA and proteins get confused as information carriers. DNA stores the recipe; proteins are the dish.

Order of operations (what to teach first)

Carbs (familiar from food), proteins (workhorse), lipids (with bilayer foreshadowing), nucleic acids (information). Save protein structure for its own segment; do not rush it.

Self-test prompts (for students before the recall deck)

- Name the monomer of each macromolecule class.
- Explain why a high fever can cause enzyme dysfunction across the body.

- Why do phospholipids form bilayers but triglycerides do not?

Why This Matters to Your Patients

Nutrition assessment, IV nutrition, lab interpretation, drug metabolism, and disease process all live here. Albumin is a protein; a low serum albumin is a marker for malnutrition or liver disease. LDL and HDL are lipid-carrying proteins. Hemoglobin A1c measures glycated hemoglobin (a sugar-protein bond), which integrates glucose exposure over months. Patients on chemotherapy lose hair because chemotherapy preferentially kills rapidly dividing cells, which require constant protein synthesis. Almost every lab value and most medications operate on macromolecules.

TOPIC 3 OF 3 . WEEK 2 . 7 CARDS (4 DOK 1 / 2 DOK 2 / 1 DOK 3)

Enzymes and ATP

How enzymes lower activation energy, what regulates them, and the role of ATP as the cell's energy currency.

The Science

Enzymes lower activation energy. They do not change the equilibrium of a reaction; they change the speed. Without enzymes, virtually no metabolic reaction would proceed fast enough to support life at body temperature. With them, reactions that would take weeks happen in milliseconds.

Specificity comes from the active site. The shape and chemistry of the active site match the substrate; this is the lock-and-key model (rigid) or, more accurately, the induced-fit model (the active site molds slightly to embrace the substrate). Anything that disrupts the active site, temperature, pH, inhibitors, allosteric regulators, changes enzyme activity. Most human enzymes have a pH optimum near 7.4 and a temperature optimum near 37 degrees. Exceptions are physiologically located: pepsin in the stomach has an optimum near pH 2.

ATP is the universal energy currency. The terminal phosphate bond is high-energy (strictly, the energy is released when ATP is hydrolyzed and the products are more stable than the reactant). ATP powers active transport, muscle contraction, biosynthesis. Cells regenerate ATP continuously through cellular respiration. A cell that runs out of ATP dies within minutes; the body has roughly enough ATP at any moment for a few seconds of activity.

Teaching This Topic

Before the video (drawing prompt)

Ask students to draw an energy diagram of a reaction with and without an enzyme. They have to predict what the curve looks like.

Common misconceptions to address

- Students think enzymes are consumed by reactions. They are not; they catalyze and are regenerated.
- Enzymes are thought to change equilibrium. They do not. They speed both directions.
- ATP is often called 'stored energy.' More accurate: ATP is the energy currency the cell uses for immediate transactions. Actual energy storage is in glycogen and fat.

Order of operations (what to teach first)

What an enzyme does (lower activation energy), why specificity exists (active site), what regulates them (temp, pH, inhibitors), then ATP and the energy-cycling story.

Self-test prompts (for students before the recall deck)

- Explain in one sentence what an enzyme does to a reaction.
- Predict the effect of pH 2 on a typical body enzyme.
- What happens to a cell that cannot regenerate ATP? Why?

Why This Matters to Your Patients

Every drug that ends in -statin, -mab, -ase, or -inib is operating on enzymes. Statins inhibit HMG-CoA reductase to lower cholesterol synthesis. SSRIs block the serotonin transporter (not technically an enzyme but a related kinetic story). Aspirin inhibits cyclooxygenase. Heart attacks release cardiac enzymes (troponin, CK-MB) that are diagnosed by enzyme assays. Organophosphate poisoning inhibits acetylcholinesterase, causing the neuromuscular crisis you may see in a farming community. Fever's danger comes from enzyme denaturation, which is why we treat fevers above 104 degrees aggressively. Understanding enzyme kinetics is understanding most of pharmacology and toxicology.